

Alzheimer's And Epilepsy: A Genetic Relationship

The Aim

The objective of the proposed project is to identify and outline a genetic relationship in terms of gene expression between Alzheimer's and epilepsy. The reason for doing so is to explain emerging evidence for the increased risk for epilepsy in Alzheimer's patients from epidemiological and observational studies. If certain genes have similar gene expressions in both these diseases and not in control groups, it can suggest that the expression of certain genes leads to the increased risk and also develop treatment methods that work in sync with both diseases.

The Background

Alzheimer's is a neurological condition which majorly affects the memory, thinking and behavioural processes (1). It's not a step in the normal process of aging, however it does gain severity with age, with the greatest risk being associated with age (1). Currently there is no cure for Alzheimer's, only treatments to help the symptoms. These treatments currently are not able to reverse the progression of the disease, but can temporarily slow down the harmful effects and symptoms (2). The exact cause of Alzheimer's is not completely known, but 70% of the risk is associated with multiple inherited genes. Epilepsy is a neurological disorder characterized by the frequent and recurrent seizures for no apparent reason (3). The cause of epilepsy is not known, but various hypotheses such as brain damage, genetic predisposition, stroke, tumours are all being studied (4). To about 70% of cases there are inexpensive anti-seizure medications available that are able to control the seizures, and for the rest other options such as surgery are available (4). Like Alzheimer's, epilepsy also develops in severity with age however there are many people that are able to control the symptoms (5). Epidemiological and observation studies have shown that Alzheimer's patients had an increased risk for developing epilepsy (5). For example, animal models have shown that in the presence of Alzheimer's, it can lead to temporal lobe epilepsy (6). A study also showed that about 10% of new-onset epilepsy in populations over 65 occurs as a result of neurodegenerative conditions (5).

The Pipeline

The basis of our pipeline is a three-step comparison (based on the premise of our comparison just being between Alzheimer's disease and Epilepsy). The first two steps are going to relate the gene expression of each diseased case with our control group. The idea is that the discrepancy in certain areas of gene expression from the healthy control group could possibly be due to the presence of the disease. This step's goal is to filter through all the genes in the human brain and isolate all of the genes that could potentially be responsible for the diseases symptoms. After finding gene expressions of significant difference, gene expressions that are similar between the Alzheimer's and epilepsy need to be found. If any genes are realized to carry this similarity, it can be used for further research into treatment targeting and expand the understanding of both the diseases. For each gene, the mean expression level of one group is compared with the mean expression level of the other, and a statistical test is performed to decide whether the two means are significantly different or in this case, not. This test will also be done in the first two steps except for statistically insignificant difference. A package from Bioconductor that runs with the R programming language, called Limma can be used. Using

these remaining similar genes, we want to then perform a final database scan for drugs affecting the gene by-products. If there are drugs after this final selection that are currently used for Alzheimer's treatment, then we can suggest that a similar treatment could also work for Epilepsy treatment, since they express these similar genes. The primary condition here is that filtered genes need to have an associated pathway or protein from our drug dataset and is found in our drug databases (SMPDB and DrugBank). This process can be done with our Limma package since the protein/drug files are in a compatible format (Comma Separated Value File). Gene files however are all in FASTA files, which can be analyzed with a Python or BASH script. With this knowledge, as mentioned we can associate drug treatments for one disease to certain genes expressed in the other diseases brains samples. For example, the use of a certain drug for Epilepsy treatment could possibly assist in new or improved Alzheimer's treatment because of the similar genes that they target.

Limma:

The main usage and function of Limma is to linear models and vector analysis in order to analyze differential expression. Vector analysis is a flexible, easy-to-use technique to compare gene expression patterns in different experimental backgrounds (7).

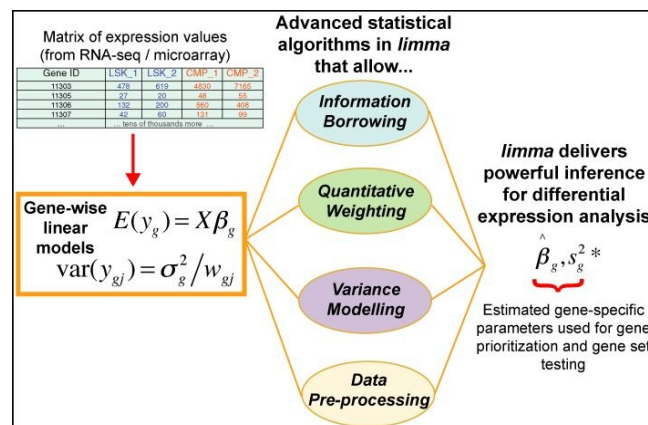


Figure 2: The process of Limma. Taking Expression values, using vector analysis, advanced statistical algorithms in order to perform differential expression analysis.

Image obtained from:

<https://www.ncbi.nlm.nih.gov/pubmed/25605792>

For each genetic probe a 2-step t-test can also be conducted, followed by Benjamini and Hochberg false discovery rate correction. This is to ensure that the results obtained are not a result of random occurrence. The p-value obtained would help to disprove the null hypothesis that the average expression level of samples in the contrast set of structures is greater than or equal to the average expression level of samples in the target set of structures.

Databases:

Allen Brain Atlas:

The Allen Brain Atlas allows users to mine through extensive gene expression data regarding adult and developing mice, humans and non-human primates (8). Human and mouse genome sequences can provide information regarding key genes required for brain development and function, as well as identifying active genes(8). In this project, the gene expression

databases for adult brains will be accessed from the Allen Brain Atlas. These databases provide downloadable microarray-based gene expression profiles for brain regions (for specific diseases or control) and colour coded expression, which is useful in making heat maps.

SMPDB/DrugBank:

The Small Molecule Pathway Database is one of the two drug databases that we are using to search for specific genes and gene pathways in humans (9). SMPDB allows us to download FASTA files for all the genes available and text files for all the protein/pathways in the database. Similarly the DrugBank database will be used in the search for drugs and drug targets of the genes we isolated (10). DrugBank allows for the downloading of the drug targeting sequence in FASTA file format and protein identifiers for the drugs in a text file. With both databases combined, a large array of possible drugs can be associated to our isolated genes and help in suggesting possible new brain disease treatment.

Genome Home Reference :

The Genome Home Reference is a general population oriented website run by the National Library of Medicine, which is part of the National Institutes of Health in the US. This will be used to narrow down the genes that will be specifically looked at in this project in order to increase feasibility and save time (11). The specific genes are: APOE, APP, PSEN1, PSEN2, GRIN2A, STXBP1, ALDH7A1, EPM2A, and NHLRC1. These were chosen based on genes most associated with Alzheimer's and Epilepsy (11).

Visualization:

Our gene expression data especially when comparing differences and similarities in the analysis can be best represented using a heat map comparing the various structures. Statistical significance can also be represented using a plotted graph with p-values.

References:

1. What is Alzheimer's? (n.d.). Retrieved from <https://www.alz.org/alzheimers-dementia/what-is-alzheimers>.
2. [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(10\)61349-9/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(10)61349-9/fulltext)
3. What is Epilepsy? (2014, January 21). Retrieved from <https://www.epilepsy.com/learn/about-epilepsy-basics/what-epilepsy>.
4. Epilepsy. WHO (n.d.). Retrieved from <https://www.who.int/en/news-room/fact-sheets/detail/epilepsy>.
5. Friedman, D., Honig, L. S., & Scarmeas, N. (2012). Seizures and epilepsy in Alzheimer's disease. *CNS neuroscience & therapeutics*, 18(4), 285-294.
6. Palop, J. J., & Mucke, L. (2010). Amyloid- β -induced neuronal dysfunction in Alzheimer's disease: from synapses toward neural networks. *Nature neuroscience*, 13(7), 812.
7. Ritchie, M. E., Phipson, B., Wu, D., Hu, Y., Law, C. W., Shi, W., & Smyth, G. K. (2015). limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic acids research*, 43(7), e47-e47.
8. Brain Map - brain-map.org. (n.d.). Retrieved from <https://portal.brain-map.org/>.
9. Alex Frolkis, Craig Knox, Emilia Lim, Timothy Jewison, Vivian Law, David D. Hau, Phillip Liu, Bijaya Gautam, Son Ly, An Chi Guo, Jianguo Xia, Yongjie Liang, Savita Shrivastava, David S. Wishart, SMPDB: The Small Molecule Pathway Database, *Nucleic Acids Research*, Volume 38, Issue suppl_1, 1 January 2010, Pages D480–D487,
10. Vivian Law, Craig Knox, Yannick Djoumbou, Tim Jewison, An Chi Guo, Yifeng Liu, Adam Maciejewski, David Arndt, Michael Wilson, Vanessa Neveu, Alexandra Tang, Geraldine Gabriel, Carol Ly, Sakina Adamjee, Zerihun T. Dame, Beomsoo Han, You Zhou, David S. Wishart, DrugBank 4.0: shedding new light on drug metabolism, *Nucleic Acids Research*, Volume 42, Issue D1, 1 January 2014, Pages D1091–D1097,

11. Genetics Home Reference - NIH. (n.d.). Retrieved from <https://ghr.nlm.nih.gov/>.